

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Lubriplate™, Part Number 3163086
- Cummins® Camshaft Removal/Installation Kit, Part Number 3164610
- Cummins® Camshaft Pilot kit, Part Number 3164845
- Camshaft installation pilot, Part Number 3164701, or equivalent.

Additional Service Items

- Lubriplate™ 105, or equivalent.

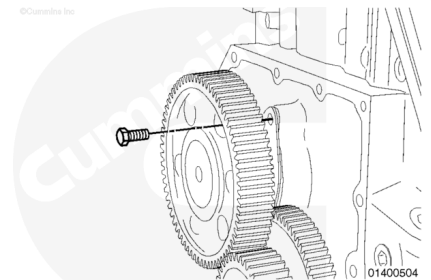
Preparatory Steps

- Engage and turn the engine barring device **counterclockwise** until the “I-6TC” mark on the damper is aligned with the pointer attached to the front of the block.
- Remove all of the rocker covers. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- For cylinders 2 and 5, identify the injector push rod which is higher in relation to the top of the rocker lever housing. If the injector push rod is higher on cylinder number 2, cylinder number 1 is on its compression stroke. If **not**, rotate the crankshaft one full revolution to place the number 1 cylinder on its compression stroke.
- Remove all the rocker assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Remove all the push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/89/89-004-014.html)
- Remove all the cam follower covers and assemblies. Refer to Procedure 004-001 in Section 4. (/qs3/pubsys2/xml/en/procedures/89/89-004-001.html)
- Drain the lubricating oil from the oil pan. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Remove the oil suction tube and cylinder block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)

- Remove the camshaft cover plate. Refer to Procedure 001-011 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-011.html>)
- Remove the flywheel housing and all related components. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/89/89-016-006.html>)

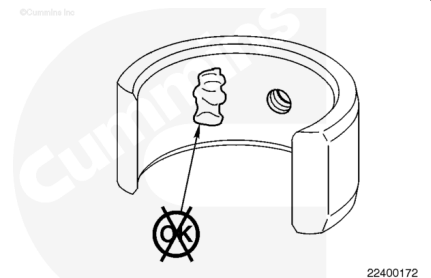
Remove

Remove the two camshaft thrust bearing mounting capscrews.



⚠ CAUTION ⚠

Make certain that the camshaft, camshaft guides, and camshaft pilot are free of lubricant and debris. Debris on the inside diameter of the camshaft guides and the camshaft lobes during removal can scratch the surface of the lobe causing performance degradation or damage to the engine.



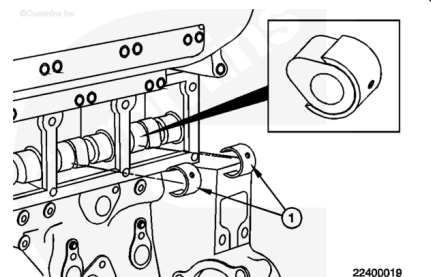
Inspect the camshaft guides for nicks and burrs, and clean thoroughly before each use. Repair or replace damaged camshaft guides.

Do **not** lubricate the inner diameter of the camshaft guides. After installing the camshaft guides, lubricate the outer diameter with Lubriplate™, Part Number 3163086.

Note : The camshaft guides are installed on the inner base circle of the camshaft lobe, not the ramp of the camshaft lobe.

⚠ CAUTION ⚠

Pilots must be used to reduce the possibility of damage to the camshaft and the bushings.



Use Cummins® Camshaft Removal/Installation Kit, Part Number 3164610, to install the C-clips **only** on the inner base circle of the valve lobes as defined and center the clips on the camshaft.

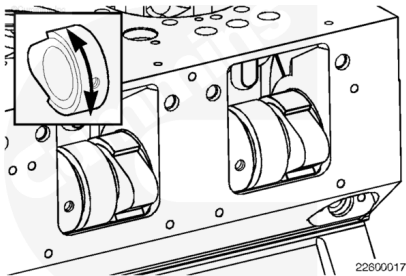
The C-clip locations can be seen in the chart below.

Camshaft Guide Installation Chart	
Camshaft Lobes	
Intake	1, 3, 5
Exhaust	4, 6

Note : The Camshaft Guide Installation Chart is for general guidance to facilitate camshaft installation; however, placement of camshaft guides can be changed to suit existing conditions.

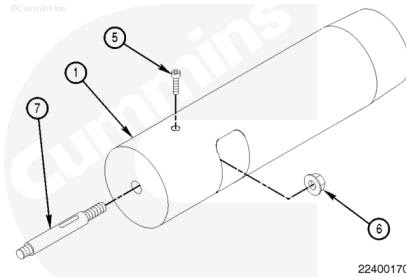
Note : It will be necessary to rotate the camshaft to install the camshaft guides. Rotate the camshaft **only** when the camshaft journals are in the camshaft bushings. Do **not** rotate the camshaft when the camshaft guides are in the bushings as this can cause the camshaft to bind.

Adjust the camshaft guides until the camshaft lobe is in the middle of the camshaft guide opening. The camshaft guides **must** be positioned flush with the camshaft lobe.



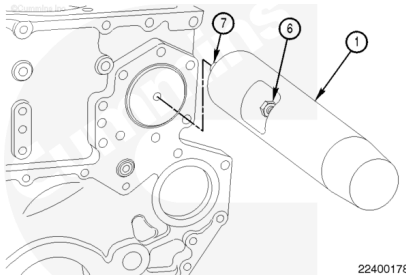
Use the Cummins® Camshaft Pilot kit, Part Number 3164845, assemble the pilot shaft (7) into the camshaft pilot (1). The end with the smaller threads goes into the pilot with the keyway oriented toward the setscrew hole.

Install the socket head capscrew (5) into the setscrew hole and hand-tighten. The pilot shaft (7) **must** be free to slide back and forth over a short distance. Thread the nut (6) onto the pilot shaft (7) through the opening in the camshaft pilot (1) and do **not** tighten.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not damage the camshaft or bushings during disassembly. Turn the camshaft gently clockwise and counterclockwise slightly while pulling the component out.

⚠ CAUTION ⚠

Rotating the camshaft too far in one direction will cause the cam lobe guides to lock up in the bushing bores.

⚠ CAUTION ⚠

Do not change the crankshaft position while the camshaft is removed.

Note : On QSK23 engines, the camshaft pilot is installed on the camshaft at the front of the engine.

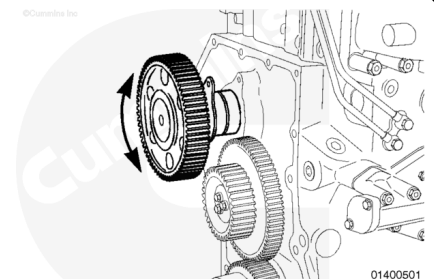
Thread the camshaft pilot (1) with the pilot shaft (7) into the threaded camshaft hole. Do **not** tighten.

Slide the camshaft pilot and camshaft assembly rearward until the front camshaft bearing journal and the camshaft pilot are in the front bearing shell. Hand-tighten the nut (6) to secure the camshaft pilot to the camshaft.

Make sure the gear teeth with the “B” marks on the cam gear and the smaller idler gear are in mesh and the keyway in the camshaft is at 6 o'clock.

Slightly rotate the camshaft back and forth during removal, if necessary.

Once the camshaft extends approximately half the length out of the block, use a sling to support the camshaft and complete the removal.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

To reduce the possibility of serious personal injury, stay clear of the camshaft when it is being raised or lowered.

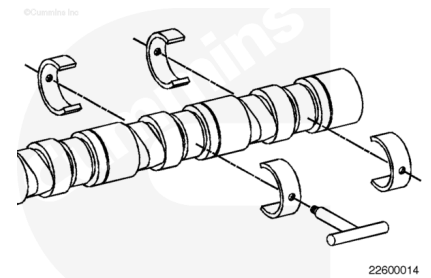
⚠ WARNING ⚠

Make sure the camshaft is balanced when lifting. If the camshaft is not balanced, it can fall and cause personal injury.

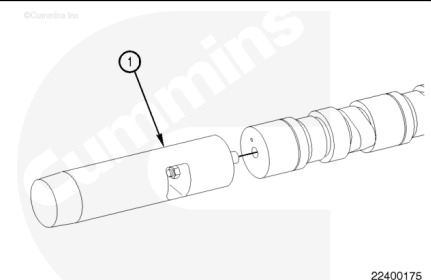
Use a lifting strap secured around the camshaft to lift the camshaft slightly and support it. Lifting the camshaft too much or too little will bind the camshaft in the bore. Carefully guide the camshaft through the camshaft bore. Make certain that the camshaft is balanced on the lifting strap.

Remove the camshaft from the cylinder block.

After removing the camshaft, use the camshaft guide remover to remove the camshaft guides from the camshaft.



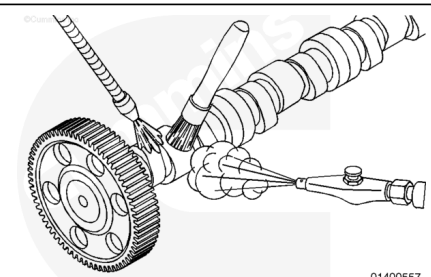
Loosen the nut and remove the camshaft pilot (1) from the camshaft.



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

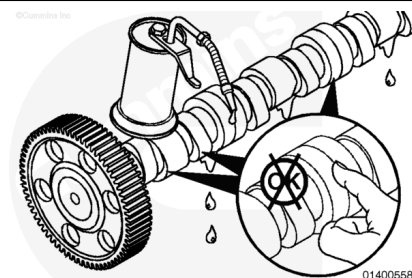


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent and compressed air to clean the camshaft.

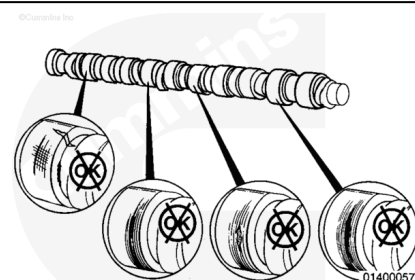
After the camshaft has been cleaned, do **not** touch the machined surfaces with bare hands. This will cause rust to form. Lubricate the camshaft with clean 15W-40 engine oil before handling.



Inspect for Reuse

⚠ CAUTION ⚠

Do not repair the camshaft by grinding the valve or the injector lobes.



Inspect the camshaft.

Check the valve and the injector lobes for damage.

If the camshaft is damaged, it **must** be replaced.

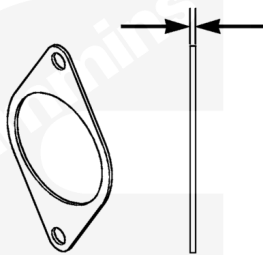
Measure each of the seven bushing journals.

Measure the outside diameter.

Camshaft Bushing Journal Outside Diameter		
mm		in
104.88	MIN	4.129
104.96	MAX	4.132

If the outside diameter is **not** within specifications, the camshaft **must** be replaced.

Inspect the camshaft thrust bearing. Refer to Procedure 001-056 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-056.html>)

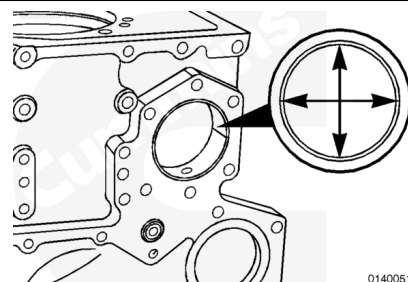


01400500

Inspect the camshaft bushings.

Measure the inside diameter.

Camshaft Bushing Inside Diameter		
mm		in
105.01	MIN	4.134
105.10	MAX	4.138

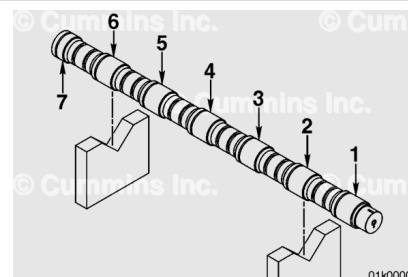


01400510

If the bushing is **not** within specifications, the bushing **must** be replaced.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



01K00003

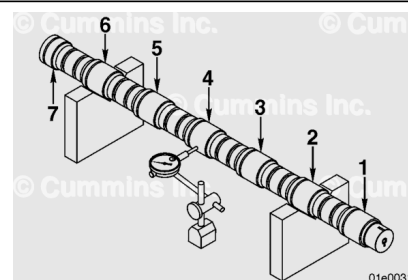
Place two vee blocks on a flat surface.

Support the camshaft on the vee blocks at journals 2 and 6.

Position a dial indicator so the stem touches the center line of the journal. Rotate the camshaft. Measure the total indicator runout for each journal.

The dial indicator **must** be positioned at the centerline of any journal measured.

Maximum Journal Runout		
mm		in
0.03	MAX	0.001



01K00011

If the camshaft is not within specification, it **must** be replaced.

Note : The maximum journal runout applies to each journal.

Note : The two supported journals do not need to be measured.

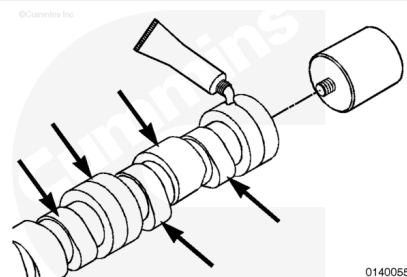
Install

Install the camshaft installation pilot, Part Number 3164835, or equivalent, on the end of the camshaft.

Thread the camshaft pilot with pilot shaft into the hole in the end of the camshaft, do **not** tighten.

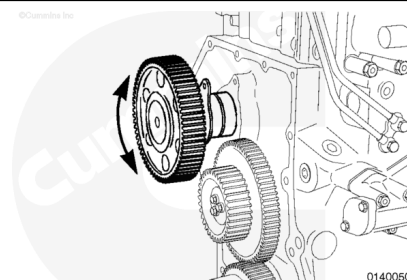
Install the camshaft guide kit, Part Number 3164610, or equivalent.

Use Lubriplate™ 105, or equivalent, to lubricate the camshaft and camshaft bushings.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



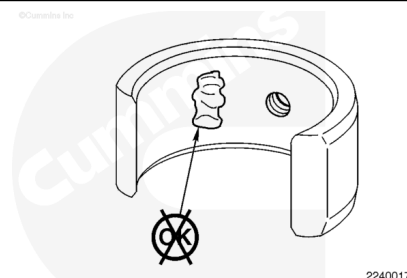
⚠ CAUTION ⚠

Be careful when installing the camshaft to reduce the possibility of the camshaft lobes damaging the bushings.

Turn the keyway in the camshaft to the 6 o'clock position.

⚠ CAUTION ⚠

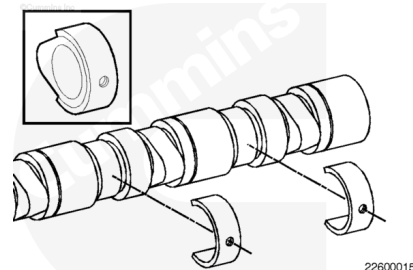
Make certain that the camshaft, camshaft guides, and camshaft pilot are free of lubricant and debris. Debris on the inside diameter of the camshaft guides and the camshaft lobes during removal can scratch the surface of the lobe causing performance degradation or damage to the engine.



Inspect the camshaft guides for nicks and burrs, and clean thoroughly before each use. Repair or replace damaged camshaft guides.

Do **not** lubricate the inner diameter of the camshaft guides.
After installing the camshaft guides, lubricate the outer diameter with Lubriplate™, Part Number 3163086.

The camshaft guides are installed on the inner base circle of the camshaft lobe, **not** the ramp of the camshaft lobe.



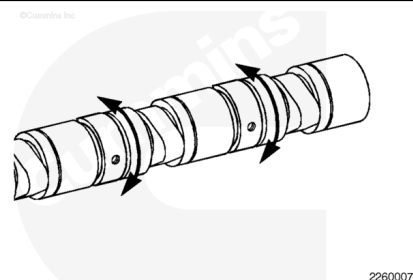
Camshaft Guide Installation Chart

Camshaft Lobes

Intake	1, 3, 5
Exhaust	4, 6

Note : The Camshaft Guide Installation Chart is for general guidance to facilitate camshaft installation; however, placement of camshaft guides can be changed to suit existing conditions.

Adjust the camshaft guides until the camshaft lobe is in the middle of the camshaft guide opening. The camshaft guides **must** be positioned flush with the camshaft journal.



⚠ WARNING ⚠

Wear gloves when handling the camshaft. The camshaft has sharp edges that can cause injury if not handled carefully.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

To reduce the possibility of serious personal injury, stay clear of the camshaft when it is being raised or lowered.

⚠ WARNING ⚠

Make sure the camshaft is balanced when lifting. If the camshaft is not balanced, it can fall and cause personal injury.

Use a lifting strap secured around the camshaft to lift it into a position parallel to the camshaft bore axis. Make certain that the camshaft is balanced on the lifting strap.

⚠ CAUTION ⚠

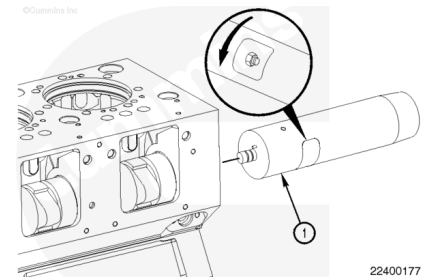
Do not rotate the camshaft during installation. The camshaft guides can bind and make installation difficult. If necessary, it is permissible to slightly wiggle the camshaft.

Carefully guide the camshaft into the camshaft bore. Lifting the camshaft too much or too little will bind the camshaft in the bore.

Slide the camshaft pilot and camshaft assembly into the rear bearing until the front bearing journal and the camshaft pilot are in the rear bearing shell. Tighten the nut (6) to secure the pilot to the camshaft.

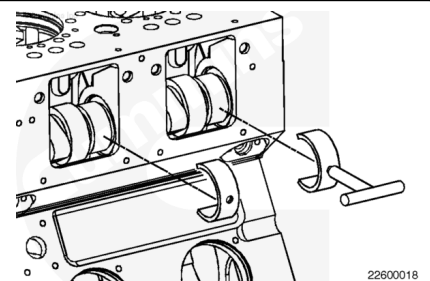
After the camshaft has been installed halfway, the lifting strap can be removed.

Once the camshaft is installed, loosen the nut and remove the camshaft pilot (1) from the camshaft.



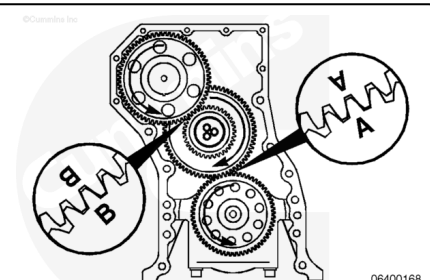
Use the camshaft guide remover to remove the camshaft guides.

It will be necessary to rotate the camshaft to remove the camshaft guides. Rotate the camshaft **only** when the camshaft journals are in the camshaft bushings. Do **not** rotate the camshaft when the camshaft guides are in the bushings as this can cause the camshaft to bind.



Align the "B" marks on the cam gear and the smaller idler gear.

After installing the gear and camshaft, double check that the "A" and "B" marks are lined up on the camshaft compound idler gear.

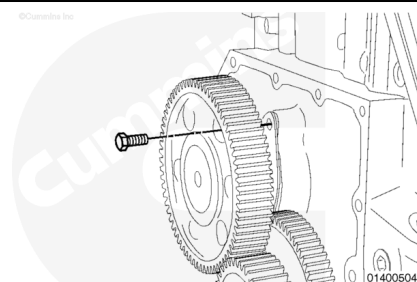


Install the two camshaft thrust bearing mounting capscrews.

Tighten the capscrews

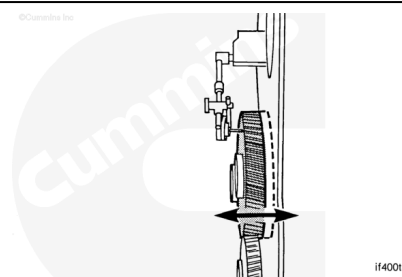
Torque Value: 64 n·m [47 ft·lb]

Remove the camshaft installation pilot and guide kit.



Use a dial indicator to measure the camshaft end clearance.

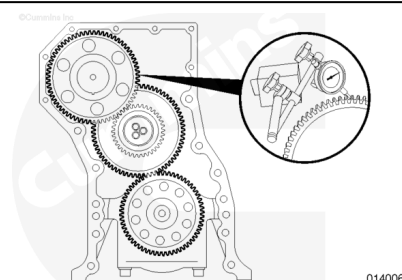
Camshaft End Clearance		
mm		in
0.05	MIN	0.002
0.20	MAX	0.008



If the clearance is **not** within specifications, check for foreign material or a piece of gasket between the thrust plate and the block.

Use a dial indicator to measure the camshaft gear backlash.

Camshaft Gear Backlash		
mm		in
0.145	MIN	0.006
0.380	MAX	0.015



Finishing Steps

- Install the flywheel housing and all related components. Refer to Procedure 001-020 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-020.html>)
- Install all the cam follower covers and assemblies. Refer to Procedure 004-001 in Section 4. (</qs3/pubsys2/xml/en/procedures/89/89-004-001.html>)
- Install the camshaft cover plate. Refer to Procedure 001-011 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-011.html>)
- Install the oil suction tube and cylinder block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>)

- Fill the lubricating oil pan. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Install all the push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/89/89-004-014.html)
- Install all the rocker assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Adjust the overhead setting. Refer to Procedure 003-006 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-006.html)
- Install all the rocker covers. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)

Last Modified: 14-Nov-2018
